s

&n);

e is %d",(n*n));

# Pattern programs:

## Right triangle pattern

```
1) int n,i,j;
   scanf("%d",&n);

   for(i=1;i<=n;i++)
   {
      for(j=1;j<=i;j++)
      {
         printf(" *");
      }
      printf("%n");
   }
```

```
*
* *
* * *
```

put printf("%d",j)

for

```
1
1 2
1 2 3
```

put printf("%d",i);

for

```
1
2 2
3 3 3
```

2) Reverse right triangle

```
int n,i,j;
scanf("%d",&n);

for (i=n;i>=0;i--)
{
    for (j=i;j>=0;j--)
    {
        printf(" *");
    }
}
```

```
* * *
* *
*
```

Put printf(" %d", i)
for

```
3 3 3
2 2
1
```

Put printf(" %d", j)
for

```
3 2 1
2 1
1
```

3) Rhombus pattern.

```
int n,i,j;
scanf("%d",&n);

for(i=1;i<=n;i++)
{
    for(j=1;j<=i;j++)
    {
Line 1      printf(" *");
    }
    printf("\n");
}
for(i=n;i>0;i--)
{
    for(j=i;j>0;j--)
    {
Line 2      printf(" *");
    }
    printf("\n");
}
```

```
*
* *
* * *
* *
*
```

Put printf("%d", i); in line 1 & line 2 for

```
1
2 2
3 3 3
2 2
1
```

Put printf("%d", j); in line 1 & line 2 for

```
1
1 2
1 2 3
1 2
1
```

4) Floyd's triangle

```
int n, i, j, s=1;
scanf("%d", &n);
for (i=1; i<=n; i++)
{
    for (j=1; j<=i; j++)
    {
        printf("%d", s);
        s++;
    }
    printf("\n");
}
```

```
1
2 3
4 5 6
7 8 9 10
```

5) Equaletral $\Delta^{le}$

```
int n, i, j, k, s;
scanf ("%d", &n);
s = n;
for (i=1; i<=n; i++)
{
    for (j=1; j<s; j++)
    {
        printf("  ");  (empty space)
    }
    for (k=1; k<=i; k++)
    {
        printf(" *");
    }
    printf("\n");
    s--;
}
```

Line 3

Line

```
    *
   * *
  * * *
 * * * *
```

Put printf("%d", i) in line 1 for

```
      1
    2   2
  3   3   3
4   4   4   4
```

printf("%d", j) for

```
      1
    1   2
  1   2   3
1   2   3   4
```

6) Inverted eqn $\Delta^{le}$

```
int n, i, j, k, s;
scanf("%d", &n);
Line 3  s = 0;
for(i=n; i>=1; i--)
{
    for(k=0; k<s; k++)
    {
        printf("  ");  (empty space)
    }
    for(j=i; j>=1; j--)
    {
Line 1      printf(" *");
    }
    printf("\n");
    s++;
}
```

```
* * * *
  * * *
    * *
      *
```

Put printf("%d", i);

in line 1 for

```
3 3 3
  2 2
    1
```

Put printf("%d", j);

for in line(1)

```
3 2 1
  2 1
    1
```

7) Rectangle pattern

```
int l, b, i, j;
scanf("%d %d", &l, &b);

for (i=1; i<=l; i++)
{
    for (j=1; j<=b; j++)
    {
        printf("*");
    }
    printf("\n");
}
```

5
3

```
* * * * *
* * * * *
* * * * *
```

8) Square pattern

4

```
int a, i, j;
scanf("%d", &a);

for (i=1; i<=a; i++)
{
    for (j=1; j<=a; j++)
    {
        printf("*");
    }
    printf("\n");
}
```

```
* * * *
* * * *
* * * *
* * * *
```

9) Hollow square

4

```
* * * *
*     *
*     *
* * * *
```

```
int a, i, j;
scanf("%d", &a);
for(i=1; i<=n; i++)
{
  for(j=1; j<=n; j++)
  {
    if((i==1)||(j==1)||(i==n)||(j==n))
        printf("*");
    else
        printf(" ");  (Empty space)
  }
  prntf("\n");
}
```

10) Hollow rectangle

```
int l,b,i,j;
scanf("%d %d",&l,&b);
for(i=1;i<=l;i++)
{
    for(j=1;j<=b;j++)
    {
        if ((i==1)||(j==1)||(i==l)||(j==b))
            printf("*");
        else
            printf(" ");  (Empty space)
    }
    printf("\n");
}
```

```
5
3
* * * * *
*       *
* * * * *
```

11) Diamond pattern

```
    *
   * *
  * * *
 * * * *
  * * *
   * *
    *
```

—

Put program 5 first
then
program 6 from line 3

12) Hour glass

```
* * * *
 * * *
  * *
   *
  * *
 * * *
* * * *
```

—

Put program 6 first
then
program 5 from line 3